Chapter 5: Diffusion Models — Score-Based SDE Framework

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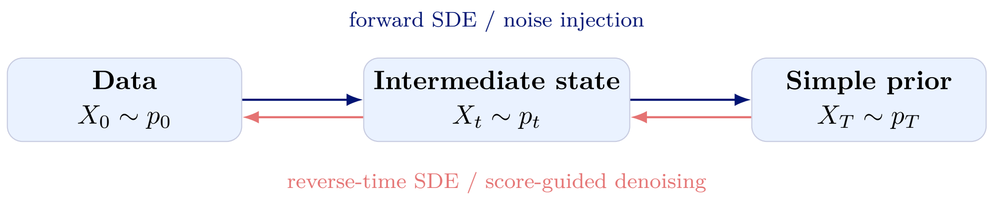
Deep Generative Models Notes



The score-SDE viewpoint places noise-conditional score networks (NCSN) and denoising diffusion probabilistic models (DDPM) inside one continuous-time framework. A fixed forward stochastic differential equation transports data to a tractable noise distribution; its time reversal transports noise back to data, provided that the time-dependent score $\nabla_x \log p_t(x)$ is known.

1 From Discrete Noise Levels to Continuous Time

1.1 The common local form

Let $0 = t_0 < t_1 < \cdots < t_L = T$ and $\Delta t = t_{i+1} - t_i$. Both discrete diffusion families have, to first order, the local update

$$X_{t+\Delta t} = X_t + f(X_t, t)\,\Delta t + g(t)\sqrt{\Delta t}\,Z_t, \qquad Z_t \sim \mathcal{N}(0, I). \tag{1}$$

Consequently,

$$p(X_{t+\Delta t} \mid X_t) \approx \mathcal{N}\big(X_t + f(X_t, t)\Delta t, g^2(t)\Delta t\, I\big)\,.$$

Equation (1) is the Euler–Maruyama discretization of

$$\mathrm{d}X_t = f(X_t, t)\,\mathrm{d}t + g(t)\,\mathrm{d}W_t, \qquad X_0 \sim p_{\text{data}}, \tag{2}$$

where W_t is a standard Wiener process and $W_{t+\Delta t} - W_t \sim \mathcal{N}(0, \Delta t\, I)$.

Remark 1: Conditioning Removes an Ambiguity

In the Gaussian transition above, X_t is the observed conditioning value, whereas $X_{t+\Delta t}$ is random. Writing both symbols as if they were free random variables obscures the meaning of the transition kernel.

1.2 NCSN / variance-exploding limit

At noise level σ_i, NCSN uses the marginal perturbation

$$X_i = X_0 + \sigma_i Z, \qquad Z \sim \mathcal{N}(0, I).$$

These marginals can be coupled as a Markov chain by injecting only the incremental variance:

$$X_i = X_{i-1} + \sqrt{\sigma_i^2 - \sigma_{i-1}^2}\, Z_i.$$

For a differentiable increasing schedule $\sigma(t)$,

$$\sigma^2(t+\Delta t) - \sigma^2(t) \approx \frac{\mathrm{d}\sigma^2(t)}{\mathrm{d}t}\Delta t.$$

Hence the continuous limit is the **variance-exploding (VE) SDE**

$$\mathrm{d}X_t = \sqrt{\frac{\mathrm{d}\sigma^2(t)}{\mathrm{d}t}}\,\mathrm{d}W_t, \qquad f(x,t) = 0. \tag{3}$$

1.3 DDPM / variance-preserving limit

For DDPM, scale the discrete noise variance as $\beta_i = \beta(t_i)\Delta t$. Then

$$X_{i+1} = \sqrt{1-\beta(t_i)\Delta t}\,X_i + \sqrt{\beta(t_i)\Delta t}\,Z_i.$$

Using $\sqrt{1-u} = 1 - \frac{1}{2}u + o(u)$ gives

$$X_{i+1} - X_i = -\frac{1}{2}\beta(t_i)X_i\Delta t + \sqrt{\beta(t_i)}\sqrt{\Delta t}\,Z_i + o(\Delta t),$$

so the limit is the **variance-preserving (VP) SDE**

$$\mathrm{d}X_t = -\frac{1}{2}\beta(t)X_t\,\mathrm{d}t + \sqrt{\beta(t)}\,\mathrm{d}W_t. \tag{4}$$

Discrete-to-continuous dictionary

Family	Discrete variance per step	Continuous coefficients
NCSN / VE	$\sigma_{i+1}^2 - \sigma_i^2$	$f = 0,\ g^2(t) = \frac{\mathrm{d}}{\mathrm{d}t}\sigma^2(t)$
DDPM / VP	$\beta_i = \beta(t_i)\Delta t$	$f(x,t) = -\frac{1}{2}\beta(t)x,\ g^2(t) = \beta(t)$

2 Forward-Time SDE: Data to Noise

Definition 2: Forward Diffusion Process

A score-based diffusion model fixes an SDE

$$\mathrm{d}X_t = f(X_t,t)\,\mathrm{d}t + g(t)\,\mathrm{d}W_t, \qquad X_0 \sim p_0 = p_{\mathrm{data}},$$

whose terminal marginal p_T is close to a tractable prior p_{prior}, usually an isotropic Gaussian.

The forward process determines two related families of densities:

- the **perturbation kernel** $p_{0t}(x_t \mid x_0)$, describing how a fixed clean sample becomes noisy;
- the **marginal density**

  $$p_t(x_t) = \int p_{0t}(x_t \mid x_0)\,p_{\mathrm{data}}(x_0)\,\mathrm{d}x_0, \tag{5}$$

  obtained after averaging over clean data.

The drift f may in principle be nonlinear. In practice, the particularly useful choice is the linear drift $f(x,t) = a(t)x$, because its perturbation kernel is Gaussian and can be sampled exactly at any time t.

3 Linear SDEs and Gaussian Perturbation Kernels

Consider

$$\mathrm{d}X_t = a(t)X_t\,\mathrm{d}t + g(t)\,\mathrm{d}W_t. \tag{6}$$

For $0 \le s \le t$, define the attenuation factor

$$A(t,s) \triangleq \exp\left(\int_s^t a(u)\,\mathrm{d}u\right).$$

Proposition 3: Gaussian Transition of a Linear SDE

Conditioned on $X_0 = x_0$, the solution of (6) is

$$X_t = A(t,0)x_0 + \int_0^t A(t,s)g(s)\,\mathrm{d}W_s.$$

Therefore

$$p_{0t}(x_t \mid x_0) = \mathcal{N}\big(x_t;\, m(t)x_0,\, v(t)I\big), \tag{7}$$

where

$$m(t) = A(t,0), \qquad v(t) = \int_0^t A^2(t,s)g^2(s)\,\mathrm{d}s. \tag{8}$$

Proof. Multiplying by the integrating factor $A(0,t)$ and applying Itô's product rule yields the stated solution. The stochastic integral is Gaussian with zero mean. Itô isometry gives its covariance as $\int_0^t A^2(t,s)g^2(s)\mathrm{d}s\, I$. □

Equation (7) also provides the direct reparameterization

$$X_t = m(t)X_0 + \sqrt{v(t)}\,\varepsilon, \qquad \varepsilon \sim \mathcal{N}(0,I). \tag{9}$$

3.1 The VE and VP cases

VE SDE. For $a(t) = 0$ and $g^2(t) = \frac{\mathrm{d}}{\mathrm{d}t}\sigma^2(t)$,

$$m(t) = 1, \qquad v(t) = \sigma^2(t) - \sigma^2(0).$$

If $\sigma(0) = 0$, then $p_{0t}(x_t \mid x_0) = \mathcal{N}(x_t; x_0, \sigma^2(t)I)$.

VP SDE. For $a(t) = -\frac{1}{2}\beta(t)$ and $g^2(t) = \beta(t)$, let

$$\alpha(t) \triangleq \exp\left(-\frac{1}{2}\int_0^t \beta(u)\,\mathrm{d}u\right).$$

Then

$$m(t) = \alpha(t), \qquad v(t) = 1 - \alpha^2(t),$$

and

$$p_{0t}(x_t \mid x_0) = \mathcal{N}\left(x_t; \alpha(t)x_0, \left(1 - \alpha^2(t)\right)I\right). \tag{10}$$

Remark 4: When Does the Terminal Law Forget the Data?

For a linear SDE, dependence on x_0 disappears as $m(T) \to 0$. In the VP case, $\alpha(T) \to 0$ gives $p_{0T}(x_T \mid x_0) \approx \mathcal{N}(0, I)$ and therefore $p_T \approx \mathcal{N}(0, I)$ after marginalizing over x_0. In the VE case the mean remains x_0, but sufficiently large noise makes the data-dependent shift negligible relative to the scale $\sigma(T)$.

4 Reverse-Time Stochastic Process

An ODE can be reversed by a time substitution. An SDE is subtler: diffusion spreads probability mass, so its time reversal must include a drift that reconstructs the changing density. That correction is exactly the score.

Theorem 5: Reverse-Time SDE

Assume the forward SDE (2) has sufficiently smooth positive marginals p_t, and g depends only on time. When time runs from T down to 0, the same family of marginals is generated by

$$\mathrm{d}X_t = \Big[f(X_t, t) - g^2(t)\nabla_x \log p_t(X_t)\Big]\mathrm{d}t + g(t)\,\mathrm{d}\overline{W}_t, \qquad \mathrm{d}t < 0, \tag{11}$$

initialized at $X_T \sim p_T$. Here $\overline{W}_t$ denotes Brownian motion with respect to the decreasing-time filtration.

The sign in (11) is easy to misread because $\mathrm{d}t < 0$. Introduce an increasing reverse clock $s = T - t$ and $Y_s = X_{T-s}$. Then the equivalent forward-in-s equation is

$$\mathrm{d}Y_s = \Big[-f(Y_s, T - s) + g^2(T - s)\nabla_y \log p_{T-s}(Y_s)\Big]\mathrm{d}s + g(T - s)\,\mathrm{d}W_s. \tag{12}$$

Now the score term visibly points toward regions of higher density.

Remark 6: Why the Reverse Process Works

The diffusion term keeps local stochastic exploration. The score-driven drift compensates for that spreading and moves trajectories toward high-density regions of the current marginal. Early in reverse time the noise scale is large and trajectories explore broadly; near the data endpoint the score term restores fine structure and trajectories concentrate around the data manifold.

4.1 Connection to Langevin dynamics

For the VE family $f = 0$. Define the time-varying temperature

$$\tau(s) \triangleq \frac{1}{2}g^2(T - s).$$

Equation (12) becomes

$$\mathrm{d}Y_s = 2\tau(s)\nabla_y \log p_{T-s}(Y_s)\,\mathrm{d}s + \sqrt{2\tau(s)}\,\mathrm{d}W_s. \tag{13}$$

This has the instantaneous form of Langevin dynamics, but its target density p_{T-s} and temperature $\tau(s)$ both evolve with time. Thus reverse diffusion can be viewed as *annealed Langevin transport*: it begins with strong exploration and gradually shifts toward accurate reconstruction.

5 Learning the Unknown Score

The reverse SDE is exact but unusable as written because the marginal score $\nabla_x \log p_t(x)$ is unknown. A neural network $s_\theta(x, t)$ is trained to approximate it at every noise level.

For the Gaussian perturbation kernel (7), the *conditional* score is available in closed form:

$$\nabla_{x_t} \log p_{0t}(x_t \mid x_0) = -\frac{x_t - m(t)x_0}{v(t)} = -\frac{\varepsilon}{\sqrt{v(t)}}. \tag{14}$$

Proposition 7: Denoising Score Matching Target

Let t be sampled from a training distribution on $(0, T]$, then sample $X_0 \sim p_{\text{data}}$ and $X_t \sim p_{0t}(\cdot \mid X_0)$. For any positive weight $\lambda(t)$, consider

$$\mathcal{L}_{\text{DSM}}(\theta) = \mathbb{E}\Big[\lambda(t)\, \|s_\theta(X_t, t) - \nabla_{x_t} \log p_{0t}(X_t \mid X_0)\|_2^2\Big]. \tag{15}$$

Its pointwise population minimizer is the marginal score $s_\theta^*(x, t) = \nabla_x \log p_t(x)$.

Proof. For fixed (x, t), the least-squares minimizer is the conditional expectation of the target given $X_t = x$. Differentiating the mixture identity (5) under the integral gives

$$\mathbb{E}[\nabla_x \log p_{0t}(X_t \mid X_0) \mid X_t = x] = \nabla_x \log p_t(x).$$

Thus the conditional score supplies an unbiased regression target for the unknown marginal score. □

Combining (9) and (14), training needs only clean data, a random time, and standard Gaussian noise:

$$X_t = m(t)X_0 + \sqrt{v(t)}\varepsilon, \qquad s_\theta(X_t, t) \approx -\frac{\varepsilon}{\sqrt{v(t)}}.$$

6 Sampling with the Learned Score

After training, replace the unknown score in (12) with s_θ. Starting from $Y_0 \sim p_{\text{prior}} \approx p_T$, Euler–Maruyama with reverse-time step Δs gives

$$Y_{k+1} = Y_k + \Big[-f(Y_k, T - s_k) + g^2(T - s_k)s_\theta(Y_k, T - s_k)\Big]\Delta s \tag{16}$$
$$+ g(T - s_k)\sqrt{\Delta s}\, Z_k, \qquad Z_k \sim \mathcal{N}(0, I). \tag{17}$$

1. Draw an initial sample from the terminal prior.
2. March from noise time T toward data time 0 using (16) (the *predictor*).
3. Optionally apply one or more Langevin correction steps at each noise level (the *corrector*).
4. Return the state at time 0 as the generated sample.

Framework in one line

$$\boxed{\begin{array}{c} \text{discrete noise schedule} \longrightarrow \text{forward SDE} \longrightarrow \{p_t\}_{t\in[0,T]} \\ \text{learn } \nabla_x \log p_t \;\Downarrow \\ \text{terminal prior} \longrightarrow \text{reverse-time SDE} \longrightarrow \text{samples} \end{array}}$$

NCSN and DDPM differ chiefly in their forward drift and diffusion. Gaussian perturbation kernels, score matching, and a reverse-time solver then determine the continuous model.